

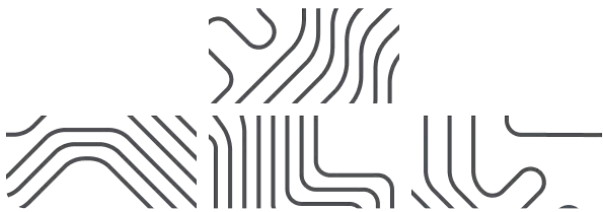


August 26, 2025

AI Server PCB Super Cycle; Copper Foil Content Upgrade

8 minutes

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PM Summary

Beneficiaries (in the preference order) :

- Copper Foil: Mitsui Kinzoku (5706:JP) and Co-Tech (8358:TT)
- Glass Fiber Cloth: Nittobo (3110:JP) and Fulltech Fiber Glass (1815:TT)
- CCL: EMC (2383:TT), TUC (6274:TT), and Doosan (000150:KR)
- PCB: GCE (2368:TT), Victory Giants (300476:CH) , WUS (002463:CH), TTM (TTMI), ISU (007660:KR), and other AI server PCB suppliers.
- PCB Drill: Union Tool (6278:JP) and Topoint (8021:TT)

Summary:

- AI server PCB content growth is a secular trend driven by quantity (layers, boards attached) and quality (material upgrades).

- AI server designs are expected to widely adopt the cableless design philosophy as it improves assembly efficiency and optimizes space.
- Copper foil is the next focus of CCL material upgrade and could face a significant supply and demand gap in 2026 if Nvidia adopts the cableless design for VR NVL144.
- Mitsui Kinzoku and Co-Tech are the main beneficiaries of the copper foil content upgrade and potential shortage.

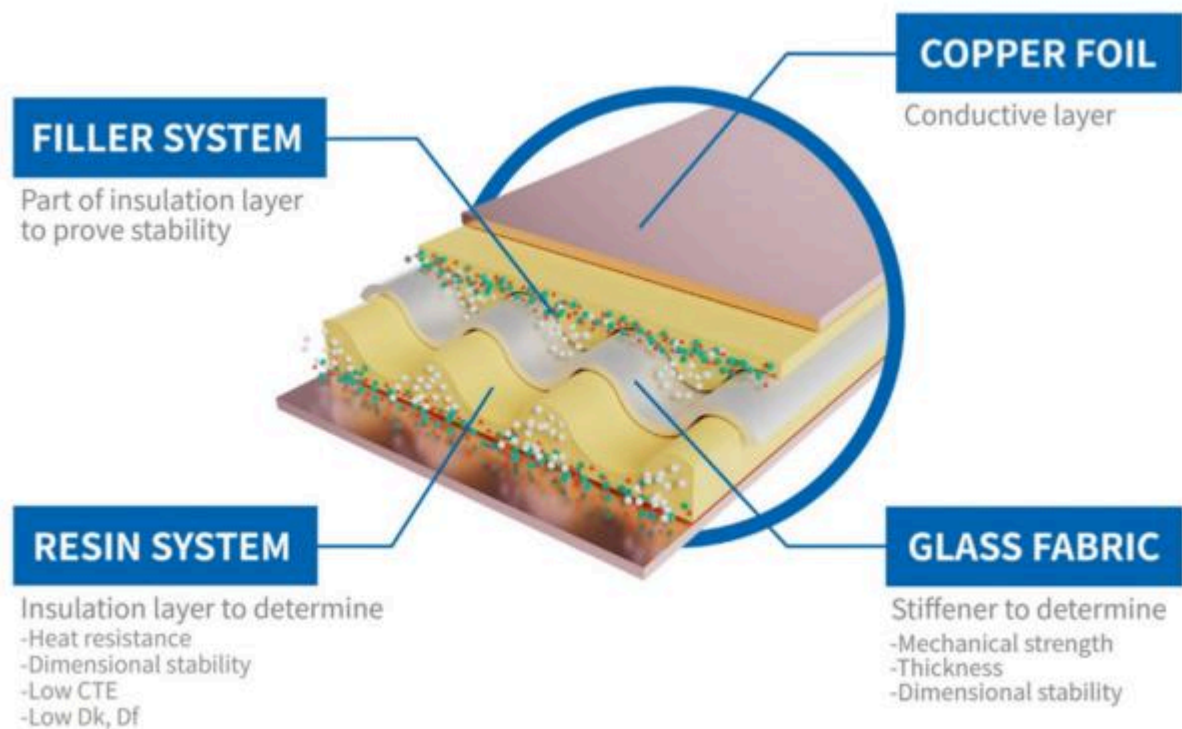
PCB Super Cycle: Content Growth Within AI Server

PCB content within an AI server continues to increase on a per accelerator basis. The key drivers of contents are the followings:

- Higher layer counts gen on gen as I/O out of the accelerator increases.
- Higher signal performance requirement leading to material upgrade.
- More board contents around accelerator: scale up switch board.
- **Cableless design to improve assembly efficiency.**

PCB layer counts of the main compute board within an AI server are increasing from 20+ layers to 30+ layers in the coming years.

The demand for higher-speed connectivity also drives significant material upgrade as signal performance is crucial in AI server. The most important component of a PCB board is the CCL, and it is made of three materials: **glass fiber cloth, copper foil, and resin**. CCL's signal performance is graded (as M7, M8, or M9) and the three materials are optimized to achieve the target performance. **For the past few years, signal performance improvement has been driven by glass fiber cloth. Going forward the material upgrade will be focused on copper foil.**

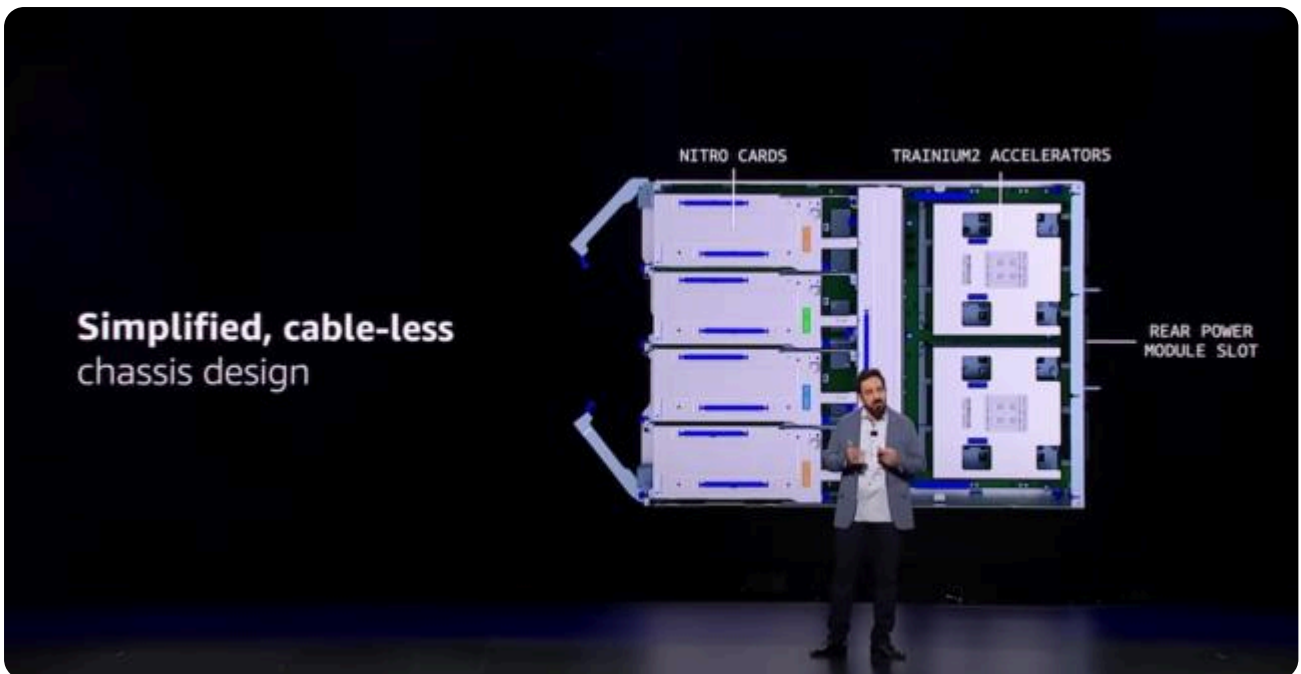


Source: Doosan

PCB Mega Trend – Reduction and Removal of Flyover Cables

As AI server assembly becomes a key bottleneck of AI server ramp, **the AI server design trends toward replacing flyover cables with PCB connectivity to improve assembly efficiency and optimize space.**

Currently, Trainium 2 is the only AI server adopting this design philosophy.



Source: AWS

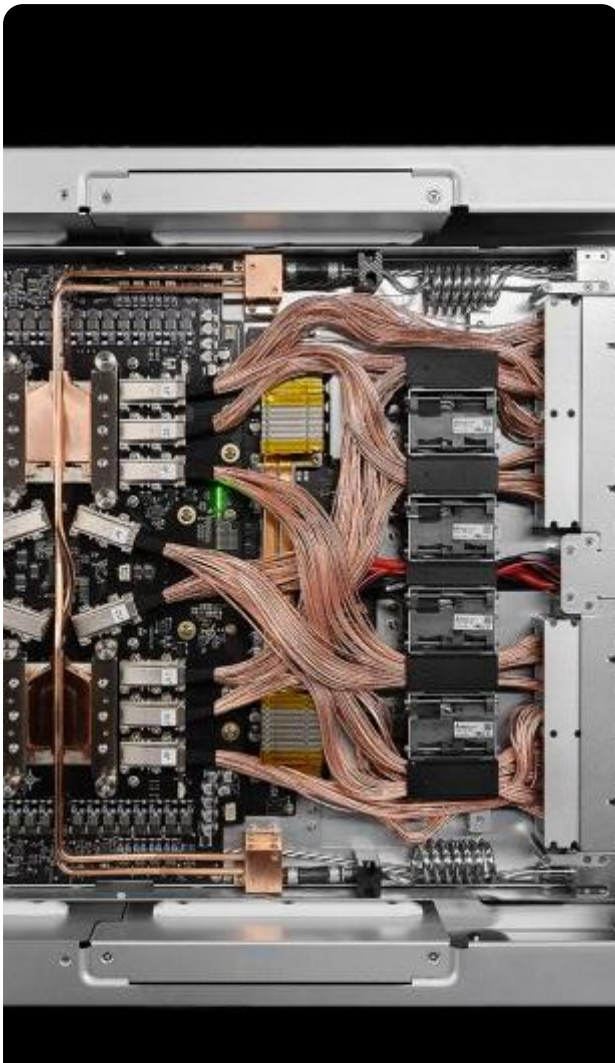
"Every cable connection is a chance for a manufacturing defect, and manufacturing defects slow you down. In fact, one of the coolest things about the Trainium 2 server is it has been designed specifically to enable automated manufacturing and assembly" – Peter Desantis, VP of AWS.

The tradeoff for more efficient assembly is the more costly PCB. As flyover cables have better signal integrity than PCB connections over the same distance, state of the art PCB materials are required to achieve the same signal performance as flyover cables. Hence, Amazon's Trainium 2 uses the highest level of CCL materials, M8 from EMC, out of all the AI servers despite having the lowest signal speed running at PCIe Gen 5.

Turning to Nvidia's Oberon platform, the ODMs encounter many GB200 assembly challenges especially with the cables **as each cable link serves as a potential point of failure.**

First, we can observe this trend from the NVSwitch tray of GB200/GB300. Initially, there are ultrapass flyover cables from the paladin connectors to the NVSwitches. At the end of last year, Nvidia decided to half the number of cables as they occupy too much space and are significant points of failure. For GB300, they took the lesson learned from GB200 and completely

removed ultra pass cables from the NVSwitch tray. The signals are all routed through the PCB now.



Source: Nvidia



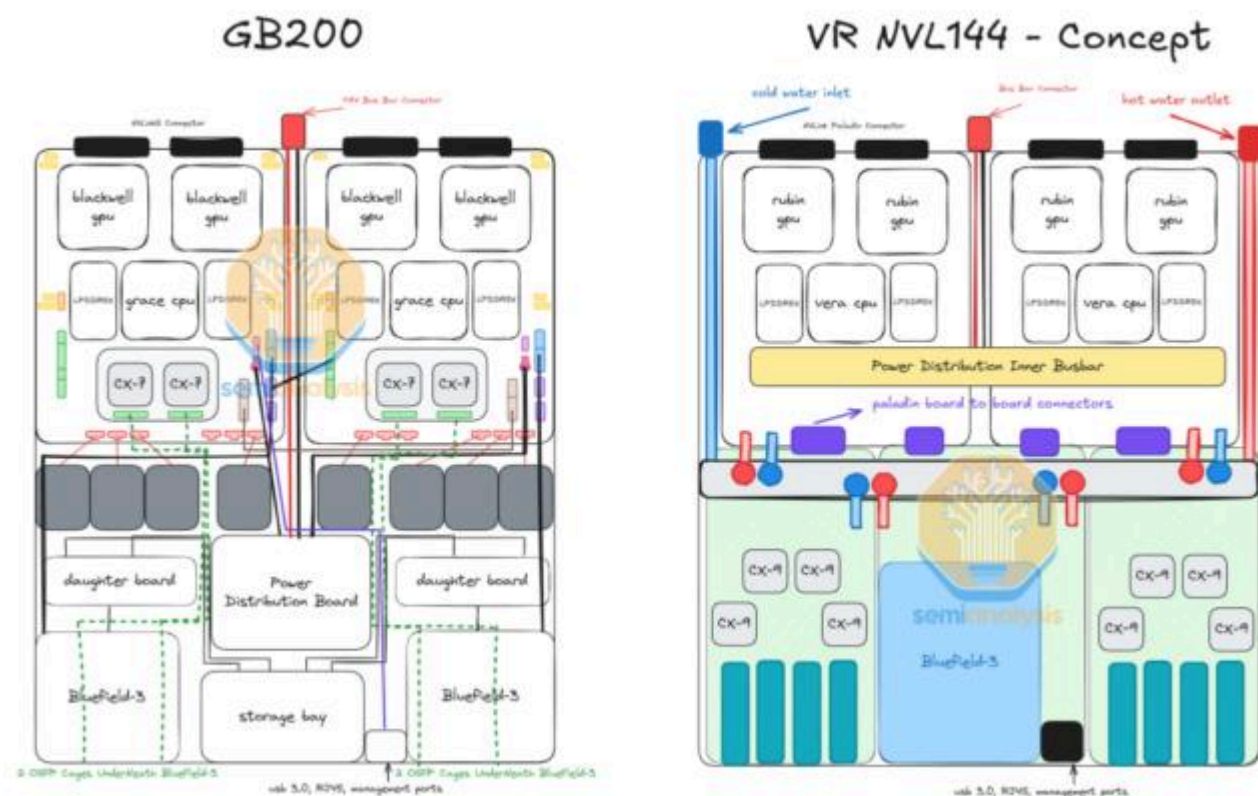
Source: SemiAnalysis, Gigabyte

We have seen the actual GB200 NVSwitch with the original design, and the ultrapass cables populate the entire chassis like hairs spread across, leaving no space for airflow.

The next critical stage of Oberon PCB content growth is the VR NVL144 system. It was too late for Nvidia to change GB200/GB300 compute tray design as it's much more complex than the NVSwitch tray. Therefore, during L10 assembly, manual laborers are required to route cables within extremely compact space, often damaging cables in the process of assembly. **For VR NVL144, we understand that Nvidia is considering the cableless design philosophy with all signals routed through the PCB just like that of the**

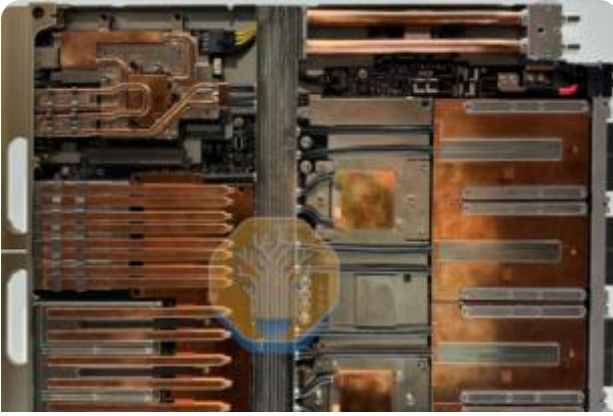
Trainium 2. Nvidia also wants to automate the assembly process, which is difficult with all the flyover cables.

(Disclaimer: While the trend indicates that the cableless design is the most ideal, the design for VR NVL144 is not finalized. Given Nvidia's track record with last minute design changes, the cableless design is not guarantee.)



Source: SemiAnalysis Estimates

For Kyber rack, the cableless design and PCB content growth trend becomes even clearer. There is no space for cables in the compute blade of Kyber rack, hence all the signals will be routed through the PCB boards. There will also be a massive PCB midplane that connects the compute blades to the NVSwitch blades.



Source: SemiAnalysis



Source: SemiAnalysis

With AWS and Nvidia's leading the trend to remove cables within an AI server, we believe the others (AMD and other AISC projects) will follow similar design philosophy as system assembly becomes increasingly complicated.

Copper Foil Upgrade

As flyover cables deliver far better signal performance than PCBs over the same distance, the cableless design means that the CCL must be upgraded to compensate for the longer signal distance over PCB. **We believe Nvidia will need to upgrade the copper foil of the CCL to HVLP 4 in VR NVL144 if it pursues a cableless design.**

Over the past two years, CCL grades in AI servers have moved from M7 to M8. This shift was mainly driven by advances in glass fiber cloth. But that material has hit a temporary wall, as the next gen quartz glass (Q-glass) supply chain is not ready for commercialization. On the contrary, copper foil hasn't reached its limit, as HVLP 4 is not widely adopted yet. Hence, **the signal performance required for VR NVL144 will be met by upgrading copper foil to HVLP 4, instead of significantly upgrading glass fiber cloth to Q-glass.**

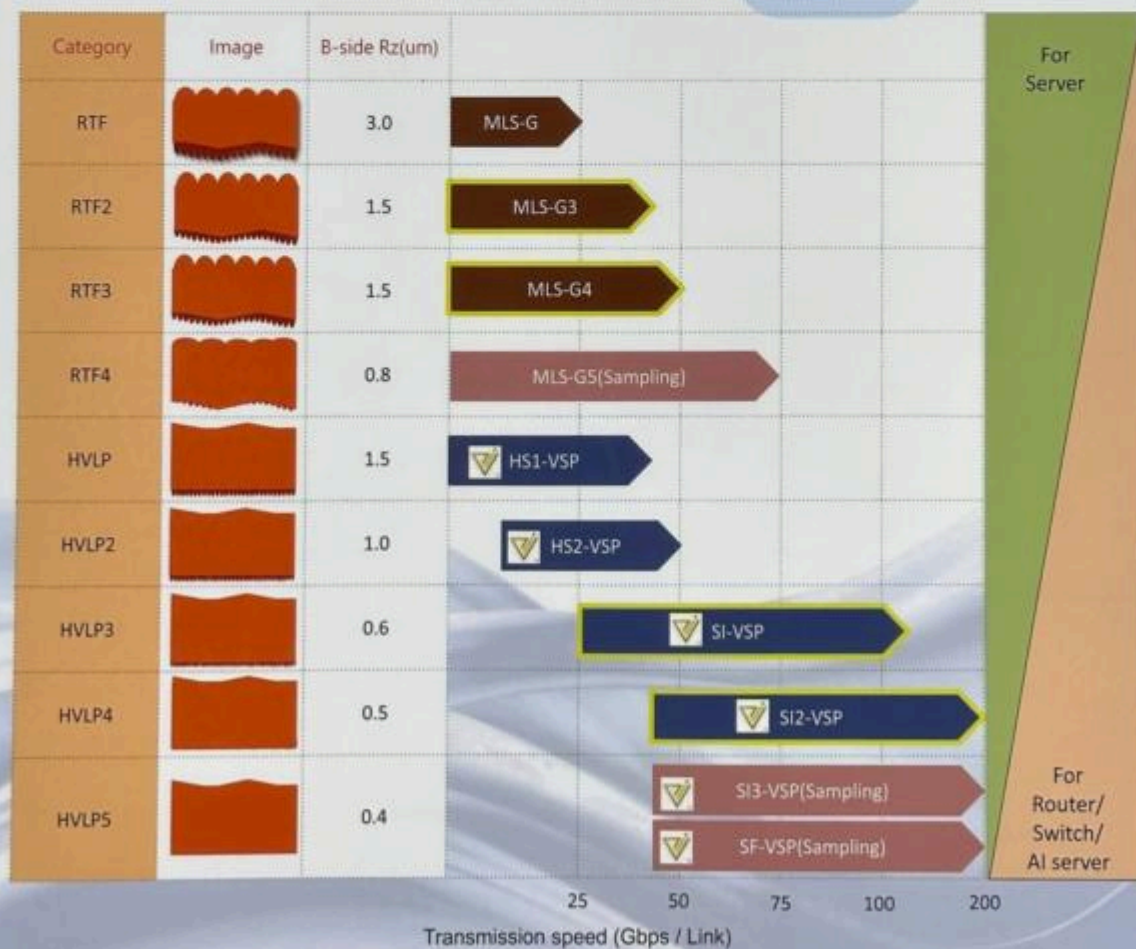
Mitsui HSD Application Foil Roadmap



高頻高速(HSD)用銅箔

Copper foil for the HSD(High Speed Digital) application

- 針對更加高頻高速的需求,發展更低耗損的銅箔材料。
- Developing copper foil that achieve sufficiently low loss to meet increasing demand for higher-speed transmission.
- 多樣化的樹脂匹配性,運用獨有的技術來達到低耗損及良好結合力的目標。
- Achieved both low loss and sufficient adhesion reliability for diversified resin systems by applying our unique technology.



*上述表列為代表性資料,非保證值。

* This is representative data not guarantee

Source: Mitsui Kinzoku, SemiAnalysis

The critical material for next gen AI server signal performance is HVLP 4 copper foil. That means the copper foil industry—especially **high-end copper foil suppliers**—will see the strongest content growth among all CCL materials benefiting from the cableless design trend.

HVLP 4 Copper Foil Supply and Demand Analysis

Given that HVLP 4 is the critical material for next generation AI server. We provide our estimate of the supply and demand of HVLP 4 copper foil in 2026.

On the demand side, the first wave of HVLP4 demand is driven by Teton 2 PDS, which is the version of Teton 2 with ALAB's PCIe scale up switch. Teton 2 PDS ramps in 2026, but the copper foil shipment leads the accelerator output by one to two quarters. Hence, we will **start to see some demand for HVLP 4 in 4Q25. We expect Teton 3 to extend the demand for HVLP 4 from AWS throughout 2026.**

The next wave of HVLP 4 demand is driven by Nvidia's VR NVL144 starting from 2Q26. In the analysis below, we modeled the demand with the assumption that Nvidia adopts the cableless design.

There is also demand from other ASICs, but the main driver of HVLP4 demand in 2026 will be from Teton 2 PDS, Teton 3 PDS/MAX and VR NVL144.

	1Q26	2Q26	3Q26	4Q26
(tons/month)				
HVLP4 Supply				
Mitsui Kinzoku	300	300	400	400
Furukawa	50	50	75	75
Co-Tech	200	200	400	400
Others	50	50	75	75
Sum	600	600	950	950
HVLP4 Demand	592	1,266	1,446	1,441
Difference	8	-666	-496	-491

Source: SemiAnalysis Estimates

Currently there are only a couple of companies supplying HVLP4 grade copper foil. The supply capacity share is currently dominated by Mitsui Kinzoku. Co-Tech comes second as it plans to shift capacity from legacy products to HVLP4.

As HVLP demand is expected to grow significantly next year, the leaders Mitsui and Co-Tech have already announced capacity (measured in tons) expansion plan for 2H2026. They are also planning to increase HVLP4 capacity by converting capacity of lower end products.

The important supply dynamic to note is that **capacity conversion is not one to one. In other words, converting HVLP 2 capacity to HVLP 4 capacity reduces capacity by 40% due to the longer production cycle.** Hence, we do not rule out the probability that demand for HVLP 4 capacity squeezes capacity of lower product leading to price hike across all product lines.

Another key characteristic of copper foil that could lead to price hike as long term inventory build is not possible is the preservation limitation of copper foil. Copper foil oxidizes after two weeks unless kept under refrigeration, which can extend its expiration to 6 months.

AWS’s demand of HVLP 4 in 2026 is confirmed and Nvidia’s VR NVL144 could double total HVLP 4 demand in 2H26. Given difficulty to significantly expand and convert capacity for HVLP 4 in the short term, **we believe there is potential for a significant supply and demand gap in 2026, which could ultimately lead to shortage and price hike.**

Mitsui Mining and Co-Tech Analysis

Mitsui Kinzoku and Co-Tech are the main beneficiaries of the potential shortage of HVLP 4 copper foil next year. Both companies have plans to expand HVLP capacity, while Co-Tech has more aggressive plan to exit the lower end products.

			2023					2024					2025 Forecast (Aug 8)					2025 Forecast (May 13)		
			Q1	Q2	Q3	Q4	Year	Q1	Q2	Q3	Q4	Year	Q1	Q2 Forecast	1st Half Forecast	2nd Half Forecast	Year Forecast	1st Half Forecast	2nd Half Forecast	Year Forecast
Copper foil	Volume of Sales	t/Month	1,300	1,490	1,600	1,550	1,490	1,680	1,650	1,680	1,600	1,650	1,750	1,950	1,850	1,840	1,840	1,700	1,670	1,690
VSP™	Volume of Sales	t/Month	200	250	280	280	250	240	390	420	430	370	510	590	550	600	580	440	470	460
MicroThin™ Total (a+b)	Volume of Sales	Index	100	98	98	105	100	114	121	102	94	108	143	155	149	131	140	146	131	138

Source: Mitsui Kinzoku

Mitsui Kinzoku:

- Raised full year FY25 guidance of the VSP (HVLP product) sales volume.
- Around **90% of operating income forecasted by the management for FY2025 is from the Engineered Materials segment, which is forecasted to grow in operating income driven by copper foil business.**
- Announced capacity expansion on 8/20 to increase VSP capacity per month from 580 tons/month to 850 tons/month. *(The capacity is based on capacity of HVLP 1, which will be much smaller if converted to capacity of HVLP 4.)*

Co-Tech:

- Exiting out of lower end products to focus on HVLP products.
- Capacity expansion can come online as early as the start of 2H26.

We believe that **the margin of HVLP 4 has been underestimated**. Our estimates suggest that the margin of HVLP 4 product is 50% or higher, and the margin could go beyond 60% during price hike cycle.

	HVLP2	HVLP3	HVLP4
(\$USD/KG)			
Cost	12	12	12
Process Fee	5	10	18
ASP	17	22	30
Gross Margin	29.4%	45.5%	60.0%

Source: SemiAnalysis Estimates

Both companies will benefit more than the others within the PCB supply chain from the increasing PCB content growth for the next two years, as high end copper foil becomes be the key driver of PCB signal performance.

Other Beneficiaries

Given the secular PCB content growth within the AI server, we believe the entire stack of PCB supply chain benefits from this trend. We believe the following stacks are poised to benefit the most.

- **High-end CCL suppliers: EMC, TUC, and Doosan**
- **AI High layer count/HDI PCB suppliers: Victory Giants, GCE, WUS, ISU, TTMI, FHT, Lincstech (GBM), and Dynamic.**
- **Glass fiber cloth suppliers: Nittobo and Fulltech**
- **PCB drill suppliers: Union Tool and Topoint**

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